Table 1: Comparison between fastText and other deep learning based methods

	Yahoo		Amazon Full		Amazon Polarity	
	Accuracy (per cent)	*Time*	*Accuracy (per cent)*	*Time*	*Accuracy (per cent)*	*Time*
char-CNN	71.2	1 day	59.5	5 days	94.5	5 days
VDCNN	73.4	2 hours	63	7 hours	95.7	7 hours
fastText	72.3	5 seconds	60.2	9 seconds	94.6	10 seconds

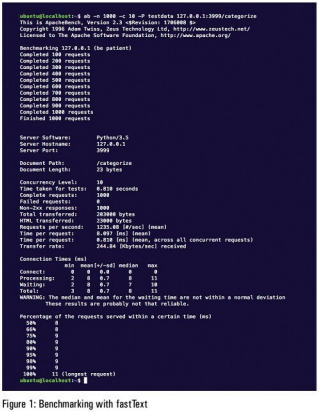

Figure 1: Benchmarking with fastText

With a new update in the fastText library, FAIR has introduced compressed text classification models which enable us to use the library even on small memory devices like mobiles and Raspberry Pi. This technique allows models using gigabytes of memory to come down to only a few hundred kilobytes, while maintaining the same performance and accuracy levels.

Now that we know how well fastText can perform, let's set it up.

Configuration and usage

It is quite simple to set up fastText. There are two ways to do this – either get the source and build it yourself, or install the Python interface for it and get started. Let's look at both methods.

Building from the source code: You will just need to get the source code from the Git repository, *https://github.com/facebookresearch/fastText.git.* Then go to the directory and enter *make*, which should compile the code and generate the executable fastText library for you. The output should be as shown in Figure 2.

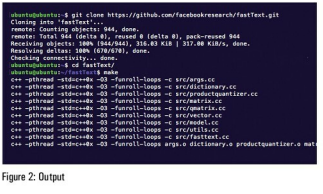

Figure 2: Output

Installation using the Python interface: This is the recommended method, as you can use it later for training and prediction purposes in the same library.

The Python module for fastText requires Cython to be installed. Execute the following commands to install Cython and fastText:

```
pip install cython
pip install fasttext
```

And you are done! Just import fastText and use the pretrained model to start predicting the classes.

```
Import fasttext
model = fasttext.load_model('model.bin')
texts = ['fastText is really amazing', 'I love fastText']
labels = classifier.predict(texts)
print labels
```

You can also refer to the Python fastText module documentation at *https://pypi.python.org/pypi/fasttext* for more details.

With the latest machine learning tools like fastText for text classification, you can certainly expect amazing products that utilise these capabilities, particularly in the field of artificial intelligence. EOU

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